

Candidate's Name:

Signature:

Random No.						Personal No.		

(Do not write your School/Centre Name or Number anywhere on this booklet.)

553/3
BIOLOGY
PRACTICAL
Paper 3
Oct. / Nov. 2022
2 hours



UGANDA NATIONAL EXAMINATIONS BOARD

Uganda Certificate of Education

BIOLOGY PRACTICAL

Paper 3

2 hours

INSTRUCTIONS TO CANDIDATES:

This paper consists of three questions.

Answer all questions.

Drawings should be made in the spaces provided.

Use sharp pencils for your drawings.

Coloured pencils or crayons should not be used.

No additional sheets of writing paper are to be inserted in this booklet.

Work on additional sheets will not be marked.

For Examiners' Use Only		
Question	Marks	Examiner's Signature & No.
1		
2		
3		
Total		

1. You are provided with specimen S and solutions X and Y, which are of different concentrations.

- (a) (i) Label **three** test tubes T_1 , T_2 and T_3 . To each of the test tubes, add a mixture of solution X and solution Y as indicated in table 1.

Table 1

Test tube	Volume of X (cm^3)	Volume of Y (cm^3)
T_1	0.0	10.0
T_2	5.0	5.0
T_3	10.0	0.0

- (ii) Using a 5 mm cork borer, obtain **three** potato cylinders from specimen S and trim each cylinder to a uniform length of 3.0 cm.
- (iii) Place **one** potato cylinder in each of the test tubes T_1 , T_2 and T_3 and ensure it is completely immersed in the solution. Keep the cylinders in the solutions for 25 minutes. (*You may continue doing other work.*)
- (iv) After 25 minutes, remove the potato cylinders from solutions and measure the final length of each of the cylinders. Record your results in table 2 and determine the changes in length of each potato cylinder.

Table 2

(03 marks)

Test tube	Initial length (cm)	Final length (cm)	Change in length (cm) - (Final length - Initial length)
T_1	3.0		
T_2	3.0		
T_3	3.0		

- (b) (i) Explain the change in length of the potato cylinder in the test tube T_1 . (04 marks)

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- (ii) Why is the change in length of the potato cylinder in test tube T_2 different from that in T_3 ? (04 marks).

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- (c) (i) Feel the potato cylinder from T_1 between your fingers, try bending without breaking it. Repeat the procedure for the potato cylinder in T_3 . Describe your observations.

Potato cylinder from T_1 (02 marks)

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Potato cylinder from T_3 (02 marks)

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- (ii) What is the significance of the observations in (c)(i) to the plant?

Potato cylinder from T_1 (02 marks)

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Potato cylinder from **T₃**

(02 marks)

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- (d) How can the observed changes in the potato cylinder in test tube **T₁** be reversed to its original state? (01 mark)

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2. You are provided with specimens **P**, **Q** and **R** which are plant organs.

- (a) Cut longitudinally through each of the specimens **Q** and **R**.

- (i) Observe the cut surfaces and state the type of fruit to which specimens **Q** and **R** belong.

Specimen **Q** (01 mark)

Specimen **R** (01 mark)

- (ii) Describe the observable structures in one half of each of the specimens **Q** and **R**.

Specimen **Q** (03 marks)

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- (iii) What are the advantages of the type of dispersal of specimen **Q** over that of **R**? (03 marks)

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- (b) Using observable features, explain how specimen **P** is dispersed. (03 marks)

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- (c) (i) Put one half of specimen **R** on a table, measure and record the length of the fruit from the stalk up to the scar. (01 mark)

Length cm

- (ii) Draw and label **one** half of specimen **R**. State the magnification of your drawing. (06 marks)

3. You are provided with specimens **T** and **U** obtained from a mammal.
- (a) Observe and feel the working surface of each of the specimens with your fingers.
- (i) Describe what you observed and felt.

Specimen **T**

(02 marks)

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Specimen U

(02 marks)

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(ii) From your observations, identify the specimens T and U.

T (01 mark)

U (01 mark)

(b) Outline **four** structural differences between specimens T and U.

(04 marks)

	T	U
(i)		
(ii)		
(iii)		
(iv)		

(c) (i) State the function of each specimen to the mammal. (02 marks)

T

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U

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- (ii) Explain **two** adaptations of specimen **U** to its function. (04 marks)

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- (d) (i) Identify the longest portion of specimen **T** and then measure it. (1/2 mark)
Record your results.

Length cm

- (ii) Draw and label specimen **T**. State your magnification. (3 1/2 marks)

Each candidate should be provided with:

1 large sized fresh Irish potato tuber, labelled S.

30 cm³ of distilled water, labelled X.

30 cm³ of solution Y.

(Solution Y is prepared by dissolving 273 g of commercial sucrose in distilled water to make 1 litre of solution.)

1 *Tridax* fruit, labelled P.

1 bean pod, labelled Q.

1 *Solanum* sp fruit (the edible type; garden egg/ bitter garden egg fruit), labelled R.

(Each of the fruits P, Q and R should have a stalk.)

1 incisor tooth, labelled T.

1 premolar tooth, labelled U.

(T and U should be obtained from the same mammal.)

1 cork borer of size 5 mm.

1 knife/scalpel.

4 test tubes.

1 stop clock.

Access to:

- Heat Source.

- Reagents for food tests.